

Dev Morwal
Indian Institute Of Technology (IIT) Bhubaneswar
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EDUCATION

Indian Institute of Technology(IIT), Bhubaneswar
B.Tech + M.Tech (Dual Degree) in Electrical Engineering

September 2020 - May 2025
Current CGPA: **7.06 / 10**

EXPERIENCE

Indian Institute of Technology, Bhubaneswar
M.Tech Research Fellow(ML, Deep Learning)

July 2024 - May 2025

- Developed new fault-detection models using Neural Networks for buck converters and 3-phase inverters to improve system reliability
- Optimized model performance for accurate fault classification under varying operating conditions
- Debugged complex issues in model predictions and improved fault detection robustness

Indian Institute Of Technology, Bhubaneswar
Research Intern(ML)

May 2024 - July 2024

- Developed a machine learning-based system for detecting gate driver faults in EV chargers using simulated grid, converter, and control unit data
- Built reusable datasets for fault scenario generation and testing Optimized system accuracy (95%) and performance across different fault conditions
- Implemented and debugged real-time monitoring and alert mechanisms for fault detection Worked for handling simulation outputs and model inputs

PROJECTS

Curvetopia : Regularization of Doodles(ML Algorithms, CNN, DSA) 

ML, DSA

- Played a key role in solving complex pattern recognition challenges by developing an AI/ML model to beautify, regularize, and auto-complete doodle drawings through detection of geometric shapes.
- Provided analytical insights by applying graph-based DFS techniques, Random Forest, and CNN models to classify curves and reconstruct structured shapes from irregular drawings.

Crack Detection (Deep Learning, TensorFlow, Semantic Segmentation) 

CNN, Python

- Played a key role in shaping automated infrastructure inspection by developing a deep learning model using UNet3++ to detect structural damage in railway infrastructure.
- Delivered analytical insights from semantic segmentation on the Tokaido Railway dataset, reducing manual inspection effort and improving damage identification accuracy.

Stock Market Prediction and Forecasting (LSTM Networks) 

Neural Network, Python

- Developed an **LSTM-based** model for stock price prediction using historical market data.
- Optimized performance through data preprocessing, hyperparameter tuning, and evaluation.
- Visualized prediction results against actual prices to assess model accuracy (92%).

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, Python, MySQL, MS Office
- **Database:** SQL-MySQL
- **Machine Learning:** Neural Networks, LLM, NLP, Computer Vision, GenAI . (Tensorflow, Keras, Pandas, NumPy, Matplotlib, Scikit-learn, LangChain, RAG)
- **Others:** Git, GitHub, Kaggle, Jupyter, LaTeX, Postman, CAD

CODING PROFILE

- **LeetCode:** https://leetcode.com/u/Dev-011_m/
- **GeeksForGeeks:** <https://www.geeksforgeeks.org/user/devmorwal261/>

RELEVANT COURSEWORK

- **Computer Science :** Data Structures and Algorithms (DSA), Object-Oriented Programming System (OOPs), Software Development Life Cycle (SDLC), Operating System (OS).
- **Mathematics:** Probability and Combinatorics, Number Theory, Set Theory

EXTRA CURRICULAR ACTIVITIES

- **Core Head of Alma Fiesta** (The Socio-Cultural Fest), IIT Bhubaneswar.
- **Led photography team** at Inter IIT Cult Meet 2023.
- **Clix, Photography Society**, IIT Bhubaneswar.
- **Secured 2nd place** in Badminton at the General Championship.